MOHAMED FATHi

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PROFESSIONAL SUMMARY

AI Agent Developer with 6 years of combined experience across software engineering and technical product management. Builds Python-based LLM applications and stateful agent workflows using LangGraph, tool calling, structured outputs, and external API integrations. Focuses on multi-agent orchestration, human-in-the-loop workflows, persistent state management, tool-level error recovery, and evidence-grounded AI outputs. Combines engineering execution with product thinking to develop practical AI automation solutions for business workflows.

CORE COMPETENCIES

- **AI Agent Development:** Python, LangGraph, LangChain, ReAct, Plan-and-Execute, Supervisor-Worker Orchestration, Tool Calling, Human-in-the-Loop (HITL), Structured Outputs.
- **LLM Integration & Search:** Groq API, OpenAI APIs, Anthropic Claude, Google Gemini, Tavily Search API, Model Context Protocol (MCP), Prompt Engineering.
- **Data & Memory:** SQLite, SQLite WAL, Persistent State, Checkpointing, SHA-256 Fingerprinting, Deduplication, Episodic Memory, Pydantic.
- **Reliability & Testing:** Tenacity, Exponential Backoff, Tool-Level Error Handling, State Validation, pytest, Failure Recovery, Process-Level Locking.
- **Backend & Engineering Tools:** FastAPI, REST APIs, Docker, Git, GitHub, CI/CD, Telegram Bot API, Make, Zapier, macOS Launchd.
- **Additional Knowledge:** Retrieval-Augmented Generation (RAG), Chroma, Pinecone, Postman, Swagger, AWS.

AI ENGINEERING PROJECTS

Agentic Job Hunter — Autonomous MENA Job Hunter Agent

Status: Open-Source Project | 2026 | Link to GitHub Repo

Python, LangGraph, Groq API, Tavily Advanced Search API, SQLite, Telegram Bot API, Tenacity, MCP, Pydantic

- *Engineered a modular LangGraph job-discovery system targeting 20 MENA job platforms and hiring feeds, combining ReAct scouting, Plan-and-Execute, Supervisor-Worker orchestration, and search-tier failover.*
- *Implemented CV-grounded cover-letter generation and self-critique, validating claims against candidate profile data and flagging unsupported or generic content before human review.*
- *Integrated human-in-the-loop approval through Telegram inline buttons and LangGraph checkpointing, enabling workflows to pause for candidate decisions and resume from persisted execution state.*
- *Designed a SQLite WAL-backed memory and deduplication layer using SHA-256 job fingerprints, incorporating human rejection feedback to suppress repeated or unsuitable opportunities.*
- *Hardened agent execution with Pydantic validation, structured tool-error handling, Tenacity-based retries, dynamic tool routing, and an MCP server separating candidate data from agent orchestration.*

EVNTOWN Outbound Engine — Freelance | Lead AI Agent Developer | 2026

Status: Production Staging | Link to GitHub Repo / Demo

Python, Groq API (LLaMA/Mixtral), Google Sheets API, IMAP, SQLite, Tenacity

- *Engineered a fully autonomous, fail-closed lead discovery pipeline utilizing targeted search queries (Tavily API) and LLM-assisted qualification to process CRM data with zero duplication.*
- *Implemented a robust state-recovery architecture using SQLite idempotency locks and IMAP Drafts-folder reconciliation to recover safely from external API or network failures.*
- *Ensured production-level reliability by integrating comprehensive system logging, observability mechanisms, and tenacity-based exponential backoff for rate limit (429/5xx) handling.*
- *-Validated system logic and state transitions to prevent unsupported AI hallucinations by restricting LLM generation to schema-validated JSON outputs injected into static HTML templates.*

Competitive Intelligence Agent (CompetiTrack)

Status: Prototype | Link to GitHub Repo

Python, LangGraph, Anthropic Claude

- *Developed an agentic research pipeline utilizing Retrieval-Augmented Generation (RAG) and Vector Databases (Pinecone/Chroma) to extract, embed, and query unstructured market data efficiently.*
- *Applied strict prompt-engineering guardrails and semantic search optimization to constrain the LLM's focus, significantly improving the fidelity of structured business reports.*

PROFESSIONAL EXPERIENCE

EVNTOWN | Remote, Egypt | Jun 2024 - Present

Technical Product Manager (Part-Time), B2B2C Mobile Marketplace

- *Owned the technical product lifecycle end to end, designing system architectures and API data flows (Draw.io, Swagger) across frontend and backend services.*
- *Managed version control and continuous integration across repositories on GitHub.*
- *Collaborated with engineering teams to optimize user acquisition funnels based on product analytics, contributing to core event conversions.*

Ready Pro Academy | Tanta, Egypt | Oct 2024 - Dec 2025

Operations Lead & Automation Architect

- *Architected automated cross-platform backend workflows with Zapier and Make, reducing manual administrative tasks.*
- *Engineered system integrations that accelerated payment cycle processing and supported improved customer satisfaction metrics.*

SAPQ Tech | Remote, Egypt | Apr 2022 - Sep 2024

Technical Product Manager, B2B Enterprise SaaS

- *Validated API endpoint reliability for the RAZ PM enterprise SaaS platform through systematic Postman testing.*
- *Monitored AWS infrastructure performance and built operational dashboards to streamline task completion tracking.*

EBTIKAR IT | Cairo, Egypt | Nov 2020 - Dec 2021

Software Engineer, Flutter / Mobile

- *Contributed to the development of SaaS mobile applications using Clean Architecture principles, managed through GitHub.*
- *Translated business requirements into technical execution, collaborating on apps that maintained high crash-free session rates.*

EDUCATION

Bachelor's Degree in Management Information Systems Obour High Institute, Egypt | Graduated: 2021

CERTIFICATIONS

- Kaggle Certifications: Python, Pandas, Intro to SQL, Intro to Machine Learning
- ChatGPT Prompt Engineering for Developers | DeepLearning.AI
- IBM Product Manager Professional Certificate | IBM via Coursera

LANGUAGES

- **Arabic:** Native
- **English:** Professional Working Proficiency